SOLUTION

SECTION–I

1. Ans. (A)

$$R = \tan\left[\lim_{x\to\infty}\sum_{r=1}^{n}\tan^{-1}\left[\frac{1+2r}{1+r^2(r+1)^2}\right]\right]$$

$$T_r = \tan^{-1}\frac{(r+1)^2-r^2}{1+r^2(r+1)^2}$$

$\therefore T_r = \tan^{-1}(r+1)^2 - \tan^{-1} r^2$

$T_1 = \tan^{-1} 2^2 - \tan^{-1} 1^2$

$T_2 = \tan^{-1}3^2 - \tan^{-1} 2^2$

$\vdots$

$T_n = \tan^{-1}(n+1)^2 - \tan^{-1} n^2$

$s = T_1 + T_2 .... T_n$

$$s = \tan^{-1}(n+1)^2 - \frac{\pi}{4}$$

$$R = \tan\left(\lim_{x\to\infty}\left\{\tan^{-1}(n+1)^2 - \frac{\pi}{4}\right\}\right) = \tan\left(\frac{\pi}{2}-\frac{\pi}{4}\right) = 1.$$

2. Ans. (D)

$$f\left(\frac{2}{x-2}\right) = \frac{1}{\frac{4}{(x-2)^2}-\frac{34}{(x-2)}+66}$$

$$= \frac{(x-2)^2}{4-34(x-2)+66(x-2)^2} = \frac{(x-2)^2}{2(3x-7)(11x-24)}$$

$\Rightarrow$ $f\left(\frac{2}{x-2}\right)$ is continuous in its domain

3. Ans. (B)

$$f(x) = \frac{\pi}{2} - \tan^{-1}\left(\frac{2x}{1-x^2}\right)$$

put $\tan^{-1}x = \alpha$ and $\alpha \in \left(0, \frac{\pi}{4}\right)$

$$\therefore \quad f(x) = \frac{\pi}{2} - 2\alpha = \frac{\pi}{2} - 2\tan^{-1}x$$

similarly, $g(x) = 2\tan^{-1}x$

$$\lim_{x\to t}\frac{f(x)-f(t)}{g(x)-g(t)} = \lim_{x\to t}\frac{\left(\frac{\pi}{2}-2\tan^{-1}x\right)-\left(\frac{\pi}{2}-2\tan^{-1}t\right)}{2\tan^{-1}x-2\tan^{-1}t}$$

$$= \lim_{x\to t}\frac{2(\tan^{-1}t-\tan^{-1}x)}{2(\tan^{-1}x-\tan^{-1}t)} = -1$$

4. Ans.(B)

$f'(x)$ =

$$\lim_{h\to0}\frac{f(x+h)-f(x)}{h} = \lim_{h\to0}\frac{f(x-h)-f(x)}{h} = -f'(x)$$

$\Rightarrow$ $f'(x) = 0$

$\Rightarrow$ $f(x) = k$ (constant)

5. Ans. (D)

Let $y = \sqrt{x+\sqrt{x-\sqrt{x+.....\infty}}}$ &

$z = \sqrt{x-\sqrt{x+\sqrt{x+.....\infty}}}$

$\Rightarrow$ $y = \sqrt{x+z}$ & $z = \sqrt{x-y}$

$\Rightarrow$ $y^2 = x+z$ & $z^2 = x-y$

$\Rightarrow$ $y^2 - z^2 = (z+y)$ $\Rightarrow$ $y - z = 1$

$\Rightarrow$ $y^2 = x + (y-1)$ $\Rightarrow$ $y = \frac{1+\sqrt{4x-3}}{2}$

Now

$$\lim_{x\to1^+}\frac{f(x)-\left(\frac{\sqrt{x}+1}{2}\right)}{(x-1)} = \lim_{x\to1^+}\frac{\sqrt{4x-3}-\sqrt{x}}{2(x-1)} = \frac{3}{4}$$

6. Ans. (B)

Sol.
$$f(x) = \begin{cases} a+(x-b)^2 & b-k \le x \le b+k \\ c+b-x & x < b-k \\ c+x-b & x > b+k \end{cases}$$

continuous at $x = b-k$ and $x = b+k$

$\Rightarrow a + k^2 = c + k$

differentiable at $x = b-k$ and $x = b+k$

$\Rightarrow -2k = -1$ and $2k = 1$

$\Rightarrow k = \frac{1}{2}$

7. Ans.(D)

$$f(1) = \frac{e+e^{-1}}{e+e^{-1}} = 1$$

$$\text{L.H.L} = \lim_{\substack{x\to1^-\\ n\to\infty}}\frac{e^{2x}+x^{2n}}{e^{2x}x^{2n}+1} = \frac{e^2+0}{1} = e^2$$

$$\text{R.H.L.} = \lim_{x\to1^+}\frac{e^{2x}+x^{2n}}{e^{2x}x^{2n}+1} = \frac{1}{e^2}$$

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8. Ans.(A,B,C)

Let $L = \lim\limits_{x\to 0}\left(\dfrac{a^x+b^x+c^x}{3}\right)^{1/x}$ (1^∞ form)

$$= e^{\lim\limits_{x\to 0}\frac{a^x+b^x+c^x-3}{3x}} = e^{\lim\limits_{x\to 0}\frac{1}{3}\left(\frac{a^x-1}{x}+\frac{b^x-1}{x}+\frac{c^x-1}{x}\right)} = (abc)^{1/3}$$

since $\dfrac{a+b+c}{3} \ge (abc)^{1/3}$ and $a+b+c=15$

$\Rightarrow$ $L \le 5$

$\therefore$ L cannot be 6.

9. Ans. B,C,D

$$f(x) = \frac{\sin\frac{\pi}{4}}{1} = \frac{1}{\sqrt{2}} \quad ; \quad 1 \le x < 2$$

$$= \frac{\sin\frac{\pi}{2}}{2} = \frac{1}{2} \quad ; \quad 2 \le x < 3$$

Hence f(x) is continuous at $\frac{3}{2}$, differentiable at $\frac{4}{3}$ & discontinuous at 2.

10. Ans. (A,C,D)

If $f(x)$ is continuous at $x = 0$

$f(0) = \lim\limits_{x\to 0} f(x)$

$$0 = \lim_{x\to 0} x^m\left(\cos x\cos\frac{1}{x}+\sin x\sin\frac{1}{x}\right)$$

$$0 = \lim_{x\to 0}\left(x^m\cos x\cos\frac{1}{x}+x^{m+1}\left(\frac{\sin x}{x}\right)\sin\frac{1}{x}\right)$$

true when $m > 0$

If $f(x)$ is differentiable at $x = 0$

$\lim\limits_{h\to 0}\dfrac{f(h)-f(0)}{h}$ exist $\Rightarrow$

$$\lim_{h\to 0}\frac{h^m\left(\cosh\cos\frac{1}{h}+\sinh\sin\frac{1}{h}\right)-0}{h} \text{ exist}$$

possible when $m > 1$

Paragraph for Question 11 to 12

11. Ans. (B)

$$P(x) = g(f(x)) = \begin{cases}(1-x)^4+(1-x)^2+1, & 1 \le x \le 2\\ (3-x)^4+(3-x)^2+1, & 2 < x \le 3\end{cases}$$

$P(2^-) = P(2^+) = 3 \Rightarrow$ P is continuous at $x = 2$

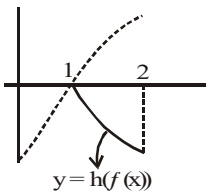
$$P'(2) = \begin{cases}4(x-1)^3+2(x-1), & 1 < x < 2\\ 4(x-3)^3+2(x-3), & 2 < x < 3\end{cases}$$

$P'(2^-) = 6$

$P'(2^+) = -6 \Rightarrow$ P is non-derivable at $x = 2$

12. Ans. (A)

$$h(f(x)) = \begin{cases}(1-x)^3, & 1 \le x \le 2\\ (3-x)^3, & 2 < x \le 3\end{cases}$$



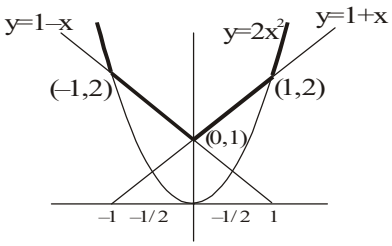
$h(f(2^-)) = -1$

$h(f(2^+)) = 1$

$\therefore$ $h(f(x))$ is discontinuous at $x = 2$

SECTION–III

1. Ans. 3



from graph

$$f(x) = \begin{cases}2x^2 & x < -1\\ 1-x & -1 \le x < 0\\ 1+x & 0 \le x < 1\\ 2x^2 & x \ge 1\end{cases}$$

hence function is continuous every where but non-differentiable at 3 points

$\Rightarrow$ $p = 0$ & $q = 3$ $\Rightarrow$ $p + q = 3$

2. Ans. 5

$|f(x)\,f(x-y)| - |f(y)\,f(x-y)| \le (x-y)^2$

$\Rightarrow |f(x-y)|\,(|f(x)| - |f(y)|) \le (x-y)^2$

$\Rightarrow f(x-y)\,(f(x)-f(y)) \le (x-y)^2$ ($\because$ $f(x) \in R^+$)

Case-I : $x > y$

$$f(x-y)\left\{\frac{f(x)-f(y)}{x-y}\right\} \le \{x-y\}$$

$$\Rightarrow \lim_{y\to x} f(x-y)\left\{\frac{f(y)-f(x)}{y-x}\right\} \le \lim_{y\to x}\{x-y\}$$

$\Rightarrow f(0).\,Lf'(x) \le 0$ ($Lf'(x)$ denotes LHD of $f(x)$)

$\Rightarrow Lf'(x) \le 0$

Case-II : $x < y$

$$f(x-y)\left\{\frac{f(x)-f(y)}{x-y}\right\} \ge \{x-y\}$$

$$\Rightarrow \lim_{y\to x} f(x-y)\left\{\frac{f(y)-f(x)}{y-x}\right\} \ge \lim_{y\to x+}\{x-y\}$$

$\Rightarrow f(0).\,Rf'(x) \ge 0$

$\Rightarrow Rf'(x) \ge 0$ ($Rf'(x)$ denotes RHD of $f(x)$)

$\because$ $f(x)$ is differentiable everywhere

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$\therefore \quad f'(x) = 0$

$\Rightarrow f(x) = \text{constant}$

$\Rightarrow f(5) = 5 \quad (\because f(0) = 5)$

3. Ans.2

$\because$ $f(x)$ is differentiable in $\left(0, \frac{1}{\sqrt{2}}\right)$

$\therefore$ $f(x)$ is continuous in $\left[0, \frac{1}{\sqrt{2}}\right]$

$\Rightarrow \quad f\left(\frac{1}{2}\right) = \lim\limits_{x \to \frac{1}{2}} f(x) \Rightarrow \lim\limits_{x \to \frac{1}{2}^-} \tan^{-1}\left(\frac{ax+b}{c}\right) = 0$

and $\lim\limits_{x \to \frac{1}{2}^+} \log\left(2 + bx^2\right) = 0$

$\Rightarrow \quad \frac{a}{2} + b = 0 \quad$ and $\quad 2 + \frac{b}{4} = 1$

$\Rightarrow \quad a + 2b = 0 \quad$ and $\quad b = -4 \quad \Rightarrow \quad a = 8$

Now, $\because f(x)$ is differentiable at $x = \frac{1}{2}$

$\Rightarrow \quad \frac{1}{1 + \left(\frac{ax+b}{c}\right)^2} \cdot \frac{a}{c}\Big|_{x=\frac{1}{2}} = \frac{1}{2 + bx^2} \cdot 2bx\Big|_{x=\frac{1}{2}}$

$\Rightarrow \quad \frac{8}{c} = \frac{-4}{2 - 4\left(\frac{1}{4}\right)} \quad \Rightarrow \quad c = -2$

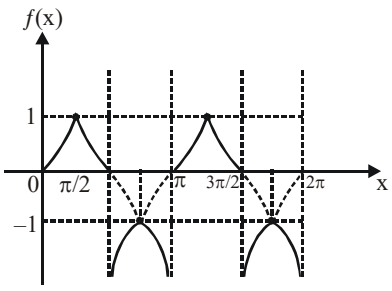
SECTION–IV

1. Ans. (A)→(r), (B)→(p), (C)→(s), (D)→(q)

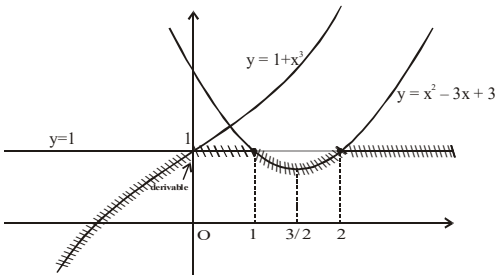
(A) $f(x) = \frac{\tan x + \cot x}{2} - \left|\frac{\tan x - \cot x}{2}\right|$

$$f(x) = \begin{cases} \cot x\ , & \tan x \geq \cot x \\ \tan x\ , & \tan x < \cot x \end{cases}$$

There are 4 points where the function is continuous but not differentiable in $(0, 2\pi)$



(B)



This hatched graph is for $f(x)$.

$\therefore f(x)$ is non-derivable at two pints

Ans = option (P)

(C) $f(x) = (x + 4)^{1/3}$

$f'(x) = \frac{1}{3}(x + 4)^{-2/3}$

Not derivable at $x = -4$

(D) $f\left(\frac{\pi}{2}\right) = \frac{\pi}{2}$

$$f\left(\frac{\pi^-}{2}\right) = \lim_{x \to \frac{\pi^-}{2}} \left(-\frac{\pi}{2}\ell n\left(\frac{2x}{\pi}\right) + \frac{\pi}{2}\right)$$

$$= \frac{\pi}{2}$$

$$f\left(\frac{\pi^+}{2}\right) = \lim_{x \to \frac{\pi^+}{2}} \left(\sin^{-1}(\sin x)\right) = \frac{\pi}{2}$$

$\Rightarrow f(x)$ is continuous at $x = \frac{\pi}{2}$

$$f(x) = \begin{cases} -\frac{\pi}{2}\ell n\left(\frac{2x}{\pi}\right) + \frac{\pi}{2} & ; \quad 0 < x \leq \frac{\pi}{2} \\ \pi - x & ; \quad \frac{\pi}{2} < x < \frac{3\pi}{2} \end{cases}$$

$$f'(x) = \begin{cases} -\frac{\pi}{2x} & ; \quad 0 < x \leq \frac{\pi}{2} \\ -1 & ; \quad \frac{\pi}{2} < x < \frac{3\pi}{2} \end{cases}$$

$$f'\left(\frac{\pi^-}{2}\right) = f'\left(\frac{\pi^+}{2}\right)$$

$\Rightarrow f(x)$ is differentiable at $x = \frac{\pi}{2}$

2. Ans. (A) → (q); (B) → (p); (C) → (s); (D) → (r)

(A) $$f(x) = \begin{cases} 1 - 1 = 0 & ; \quad 1 < x \leq 2 \\ 0 & ; \quad x = 1 \\ 1 - x & ; \quad 0 \leq x < 1 \\ -\sin \pi x & ; \quad -1 \leq x < 0 \end{cases}$$

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at x = 0, f(x) is not continuous & not differentiable

at x = 1, f(x) is continuous & not differentiable

at x = 2 and – 1, f(x) is continuous & differentiable

(B) LHD = $f'(0^-) = \lim_{h\to 0} \frac{f(0-h)-f(0)}{-h}$

$= \lim_{h\to 0} \frac{h^2 e^{-\frac{1}{h}} - 0}{-h}$

$= \lim_{h\to 0} \left(-h.e^{-\frac{1}{h}} \right) = 0$

Ans. = Option (P)

(C) $f(x) = \frac{x}{x+1}$, not defined at x = – 1

$g(x) = \frac{f(x)}{f(x)+2}$

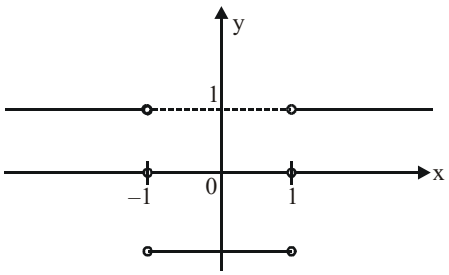
g(x) is not defined at f(x) = – 2

$\frac{x}{x+1} = -2 \quad \Rightarrow x = \frac{-2}{3}$

Also x = 0 is not in the domain of f(x)

So, at 3 points g(x) is not differentiable.

(D) $y = \begin{cases} 0 & ; \quad x = \pm 1 \\ -1 & ; \quad |x| < 1 \\ 1 & ; \quad |x| > 1 \end{cases}$



not continuous at x = 1,–1

$\therefore$ not differentiable at x = 1,–1

so no tangent exist at x = 1,–1(2 Points)

Ans. = option (r)

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